

Samiul Alam



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EDUCATION

Utah State University, USA

- Doctoral Student in Mechanical Engineering (current)
- Master of Science in Mechanical Engineering (2025)

Khulna University of Engineering and Technology, Bangladesh

- Bachelor of Science in Mechanical Engineering (2022)

HONOR/ AWARDS /GRANTS

- **College of Engineering and School of Graduate Studies** travel grant, Utah State University, 2026.
- **Runner-up in University Research Symposium**, Society for the Advancement of Material and Process Engineering (SAMPE) – North America, in MS student research category, 2025.
- **SAMPE** travel grant, 2025.
- **Graduate Research and Creative Opportunity** (GRCO) Fellowship, Utah State University, 2025.

RESEARCH EXPERIENCE

Research Focus

- **Additive manufacturing of polymer composites for automotive and aerospace** applications
- **Mechanical and Microstructural** characterization, **process-structure-property** analysis of composites

Research Projects

- Cost-effective Circular Manufacturing of Lightweight Vehicle Shells and Battery Casing by Recyclable-by-Design Polymer Composites, **DOE-Vehicle Technology Office**, 2024-present.
- Towards the Design of Advanced Protection Materials for Electrified Pavement Systems: Focusing on Electrical and Thermal Management, **NSF-ASPIRE**, 2025-2026.
- Energy Efficient, Rapid Manufacturing of Urban Air Mobility Composite Propellers Using Lamination-Based Additive Manufacturing, **Hyundai NGV**, 2024-2025.
- Performance Evaluation of Protection Materials for Roadway Embedded Wireless Charging Systems, **NSF-ASPIRE**, 2023-24.

Technical Proficiency

- Additive Manufacturing: Material Extrusion, Sheet Lamination
- Conventional Manufacturing: Resin Infusion, Vacuum Bagging, Hand Lay-up
- Mechanical and Thermal Property, Surface, and Microstructure Characterization of Composites

PUBLICATIONS

Journal Articles

1. **Alam, S.**, Young, D., Sung, D.H., and Lee, J. "Electrophoretic Deposition of Graphite Oxide on Additively Manufactured Porous Non-Conductive Thermoplastic Composites", *SAMPE Journal* 62 (3), 2026. <https://doi.org/10.33599/SJ.v62no3.01>

PUBLICATIONS

Journal Articles

2. **Alam, S.**, and Lee, J. "Performance Evaluation of Thermoplastic Composite Propellers Fabricated using Composite-Based Additive Manufacturing", *Composite Structures* 387, 2026. <https://doi.org/10.1016/j.compstruct.2026.120344>
3. Shah, S., Din, I., Hussain, M. Z., **Alam, S.**, Hassan, M.T., Altaf, K., and Lee, J. "Quasi-Static Tensile and Compression Performance and Failure Mechanisms of 3D Printed Short Fibre-Reinforced Lattice Structures", *Journal of Thermoplastic Composite Materials*, 2026. <https://doi.org/10.1177/08927057261489794>
4. Park, S., **Alam, S.**, Stoker, G., Segura, K., Seo, W. G., Young, D., Lee, J., and Sung, D. H., "Composite-Based Additive Manufacturing (CBAM): Mechanical Characterization of High-throughput Thermoplastic Composites and Demonstration for Urban Air Mobility Application", *Composites Part B: Engineering*, 2025. <https://doi.org/10.1016/j.compositesb.2025.113035>
5. Lee, D., **Alam, S.**, Yang, S.B. et al. "Control of Anti-/De-Icing Behavior and Surface Stability of MWCNT/Polyurethane Nanocomposite Coating", *Colloids and Surfaces A: Physicochemical and Engineering Aspects* 725 (2), 2025. <https://doi.org/10.1016/j.colsurfa.2025.137692>
6. **Alam, S.**, Young, D., Sung, D.H., and Lee, J. "Microstructural Characterization and Porosity Analysis in Sheet Lamination and Material Extrusion-Based Additively Manufactured Thermoplastic Composites", *SAMPE Journal* 61 (4), 2025. <https://doi.org/10.33599/SJ.v61no4.01>
7. **Alam, S.**, Hassan, M.T., Merrell, J., and Lee, J. "Comparative Analysis of Water-induced Response in 3D-printed SCF/ABS Composites under Controlled Diffusion", *SAMPE Journal* 60 (4), 2024. <https://doi.org/10.33599/SJ.v60no4.02>
8. Kamruzzaman, M., **Alam, S.**, "Influence of Stacking Sequence on the Mechanical Properties of Banana-Glass Fiber Hybrid Laminates for Automotive Shells", *Heliyon* 10 (21), 2024. <https://doi.org/10.1016/j.heliyon.2024.e40130>
9. **Alam, S.**, Bari, M.A., Nabil, S.T. et al. "Effect of pre-hydrolysis and salt- impregnation on biochar production from jute stick", *Biomass Conversion and Biorefinery* 14 (15), 2024. <https://doi.org/10.1007/s13399-023-03846-1>

Technical Conference Proceedings

1. **Alam, S.**, Segura, K. Stoker, G., Sung, D.H., and Lee, J. "Composite-Based Additive Manufacturing (CBAM): Tailoring Mechanical Properties via Thermal Consolidation Parameters", in the Proceedings of the *ASME Aerospace Structures, Structural Dynamics, and Materials (SSDM) 2026 Conference*, Long Beach, CA, June 2026.
2. Hassan, M.T., **Alam, S.**, and Lee, J., "Multiscale Prediction of Plain-Woven Composite Properties Containing Manufacturing-Induced Defects", in the Proceedings of the *ASME Aerospace Structures, Structural Dynamics, and Materials (SSDM) 2026 Conference*, Long Beach, CA, June 2026.
3. Nielsen, D., **Alam, S.**, Ricks, T.M., and Lee, J., "Multiscale Modeling and Experimental Validation of Carbon Fiber-Reinforced Pyrolytic Graphite Sheet Composites for Space Radiator Applications", in the Proceedings of the *ASME Aerospace Structures, Structural Dynamics, and Materials (SSDM) 2026 Conference*, Long Beach, CA, June 2026.
4. **Alam, S.**, Young, D., Sung, D.H., and Lee, J. "Electrophoretic Deposition on Additively Manufactured Porous Non-Conductive Thermoplastic Composites", in the Proceedings of the *Society for the Advancement of Material and Process Engineering (SAMPE) – North America*, Seattle, WA, April 2026.
5. Hassan, M.T., **Alam, S.**, and Lee, J. "Microstructure-Based Homogenization of Elastic Properties in Additively Manufactured Thermoplastic Composites", in the Proceedings of the *Society for the Advancement of Material and Process Engineering (SAMPE) – North America*, Indianapolis, IN, May 2025.

Technical Conference Proceedings

6. **Alam, S.**, Young, D., Sung, D.H., and Lee, J. "Microstructural Characterization of Sheet Lamination-Based Additively Manufactured Fiber-Reinforced Thermoplastic Composites", in the Proceedings of the *Society for the Advancement of Material and Process Engineering (SAMPE) – North America*, Indianapolis, IN, May 2025. <https://doi.org/10.33599/nasampe/s.25.0034>
7. Hassan, M.T., **Alam, S.**, and Lee, J., "A Comprehensive Study on Long-Term Durability of Protective Epoxy Coatings for Electrified Roadways," in the Proceedings of the *American Society for Composites 39th Annual Technical Conference*, San Diego, CA, October 2024.
8. **Alam, S.**, Hassan, M.T., Merrell, J., and Lee, J. "Comparative Analysis of Water-induced Response in 3D-printed SCF/ABS Composites under Controlled Diffusion", in the Proceedings of the *Society for the Advancement of Material and Process Engineering (SAMPE) – North America*, Long Beach, CA, May 2024. <https://doi.org/10.33599/nasampe/s.24.0075>
9. **Alam S.**, Robbany M.S., Bari M.A.A., and Islam M.S., "Three Stage Carbonisation: Biochar Production from Jute Stick", in the Proceedings of the *International Conference on Mechanical, Industrial and Energy Engineering*, Bangladesh, December 2022.
10. Khan M.S.M.N., **Alam S.**, and Bari M.A.A., "Numerical Analysis of Thermal Behavior of 3D Printing Nozzle Arrangement", in the *Proceedings of the International Conference on Mechanical Engineering and Applied Sciences*, Bangladesh, December 2022.
11. Alam M.M., Rahman R., Kawsar A.K.A., Ahmed H., Uday S.I., **Alam S.**, and Tasnim T., "Numerical Investigation of Heat Transfer Characteristics of Aluminium and Copper Heat Spreader", in the Proceedings of the *International Conference on Mechanical Engineering and Renewable Energy*, Bangladesh, December 2021.
12. Alam M.M., Ahmed I., Joy M.G.R., Kamal M., Anim M.T., Islam T., Amin I., and **Alam S.**, "Development and Performance Optimization of Vertical Axis Wind Turbine Using AVP", in the Proceedings of the *International Conference on Mechanical Engineering and Renewable Energy*, Bangladesh, December 2021.

Poster/Presentation

1. **Alam, S.** and Lee, J., "Tailoring Failure Mechanisms via Post-Processing Parameters in Composite-Based Additive Manufacturing (CBAM)," *ESIA18 – ISSI2026 International Conference*, Glasgow, UK, May 2026.
2. **Alam, S.**, Hassan, M.T., and Lee, J. "Microstructure-to-Property Relationship of Thermoplastic Composites Fabricated via Composite-Based Additive Manufacturing Process", *SAMPE Conference*, Indianapolis, IN, May 2025.
3. **Alam, S.** and Lee, J. "Process-dependent Microstructural Variations in Sheet Lamination (SHL)-Manufactured Fiber-reinforced Thermoplastic Composites", *SAMPE Utah Chapter*, Logan, UT, February 2025.
4. Hassan, M.T., **Alam, S.**, and Lee, J., "Performance Evaluation of Epoxy Coatings for Electrified Roadways under Environmental and Mechanical Impacts", *ASPIRE Annual Meeting & Technology Showcase*, Logan, UT, September 2024.
5. **Alam, S.**, Hassan, M.T., Merrell, J. and Lee, J., "Understanding Water Diffusion Characteristics in Civil-Grade Epoxy Materials: A Comprehensive Study Through Microscopic Observation and Numerical Analysis", *SAMPE Conference*, Long Beach, CA, May 2024.